MATHS MANTHAN
• TUITION CLASSES •
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**Subject:** Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
**Class:** Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

## SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

**Q1.** If the mean and median of a set of data are 24 and 26 respectively, then its mode is: [1]
(A) 28 (B) 30 (C) 32 (D) 25

**Q2.** For the following distribution, the cumulative frequency of the class $20 - 30$ is: [1]

| Class | 0–10 | 10–20 | 20–30 | 30–40 |
|---|---|---|---|---|
| Frequency | 5 | 8 | 12 | 6 |

(A) 12 (B) 13 (C) 25 (D) 31

**Q3.** The algebraic sum of the deviations of a set of values $x_1, x_2, \ldots, x_n$ from their mean $\bar{x}$ is always: [1]
(A) 1 (B) 0 (C) $> 0$ (D) $< 0$

**Q4.** If the median of the observations $11, 12, 14, 18, x+2, x+4, 30, 32, 35, 41$ arranged in ascending order is 24, then $x$ is: [1]
(A) 21 (B) 22 (C) 20 (D) 24

**Q5.** **Assertion (A):** If the difference between mode and median of a frequency distribution is 24, then the difference between median and mean is 12.
**Reason (R):** For any moderately skewed frequency distribution, Mode − Mean = 3(Median − Mean). [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

**Q6.** If the mean of five observations $x, x+2, x+4, x+6, x+8$ is 11, find the value of $x$. [2]

## SECTION C: Short Answer Type (3 × 3 = 9 Marks)

**Q7.** Find the mean of the following distribution using the Assumed Mean Method (take $a = 25$): [3]

| Class Interval | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 |
|---|---|---|---|---|---|
| Frequency | 5 | 12 | 20 | 9 | 4 |

**Q8.** Find the missing frequency $p$ if the mean of the following distribution is 18: [3]

| Class | 11–13 | 13–15 | 15–17 | 17–19 | 19–21 | 21–23 | 23–25 |
|---|---|---|---|---|---|---|---|
| Frequency | 7 | 6 | 9 | 13 | $p$ | 5 | 4 |

**Q9.** Calculate the median for the following distribution of scores: [3]

| Score | 50–60 | 60–70 | 70–80 | 80–90 | 90–100 |
|---|---|---|---|---|---|
| Frequency | 12 | 14 | 8 | 6 | 10 |

## SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

**Q10. CASE STUDY: RESIDENTIAL ELECTRICITY ENERGY CONSUMPTION SURVEY** [4 Marks]

A city electricity distribution corporation audited the weekly power consumption of 100 residential households in a suburban sector to evaluate energy demand under peak summer loads.

| **Consumption (Units)** | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 |
|---|---|---|---|---|---|---|---|
| **Number of Families** | 16 | 12 | 18 | 20 | 16 | 10 | 8 |

(i) Find the median class for the electricity consumption of the families. [1]

(ii) Calculate the median weekly power consumption of the families. [2]

(iii) What percentage of families consumed 40 units or more? [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★